CSE4203: Computer Graphics

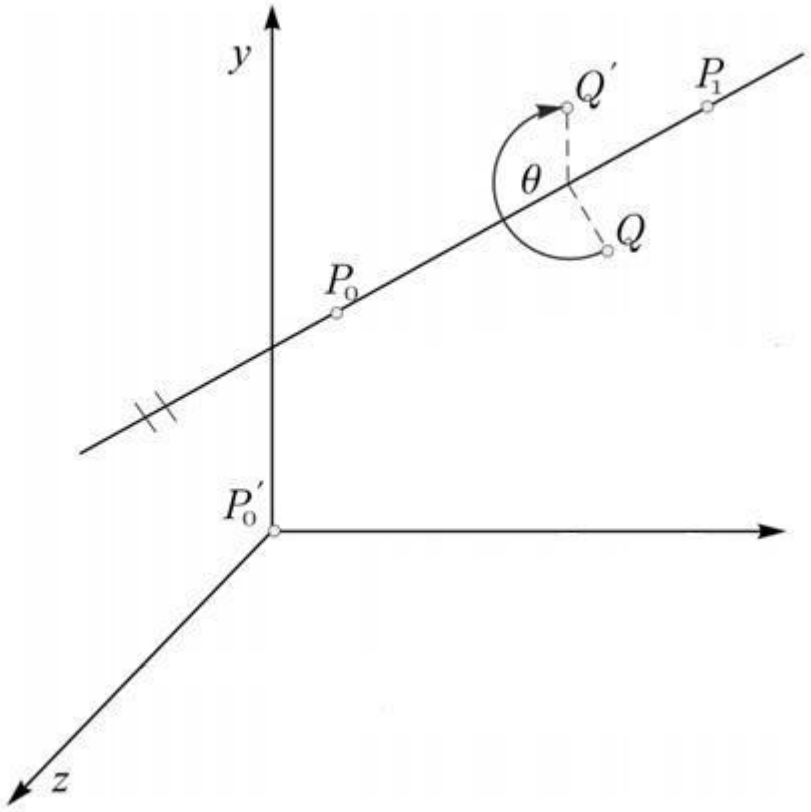
Lecture – 4 (part - C)

Transformation Matrices

Outline

- 3D Transformation
- Rotation about an arbitrary line

Rotation about an arbitrary line (1/1)

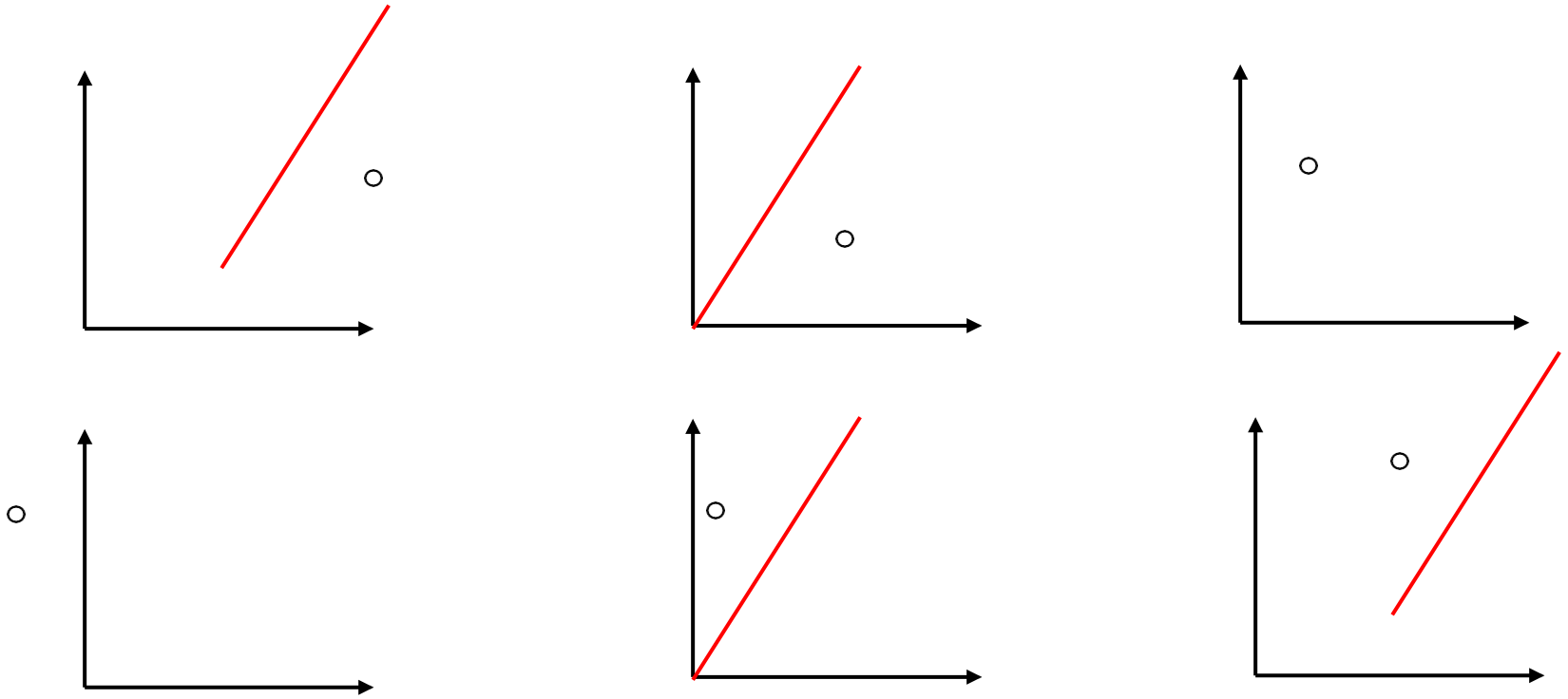


The basic idea is to make the arbitrary rotation axis coincide with one of the principle axis. Assume an arbitrary axis in space passing through the point **P0 (x0, y0, z0)** and **P1 (x1, y1, z1)** .

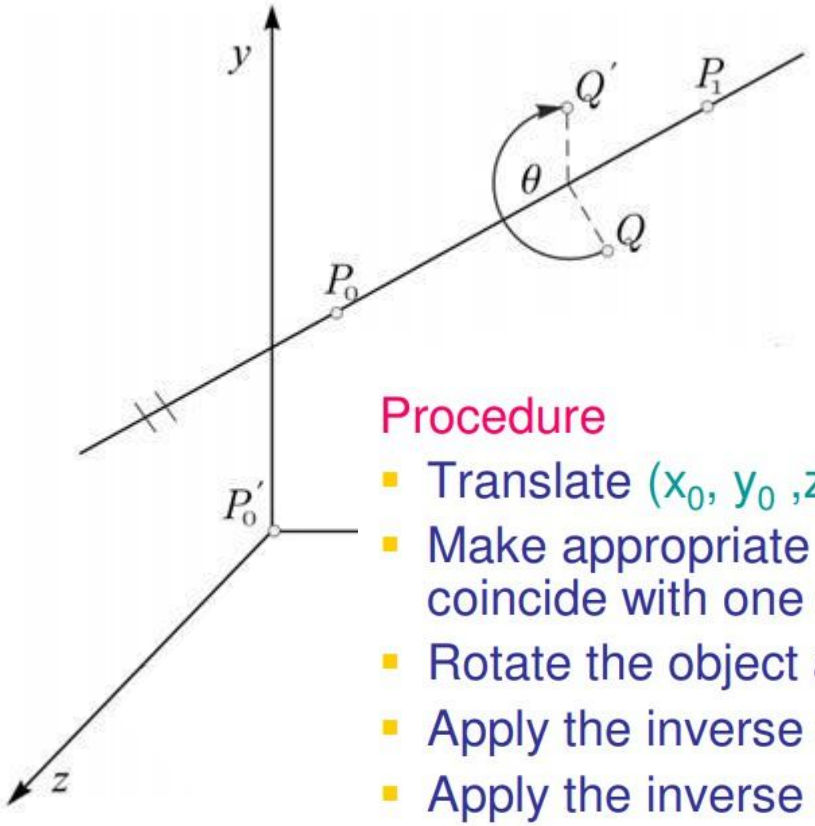
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In 2D case (1/1)

Reflecting about an arbitrary line



Steps (1/1)

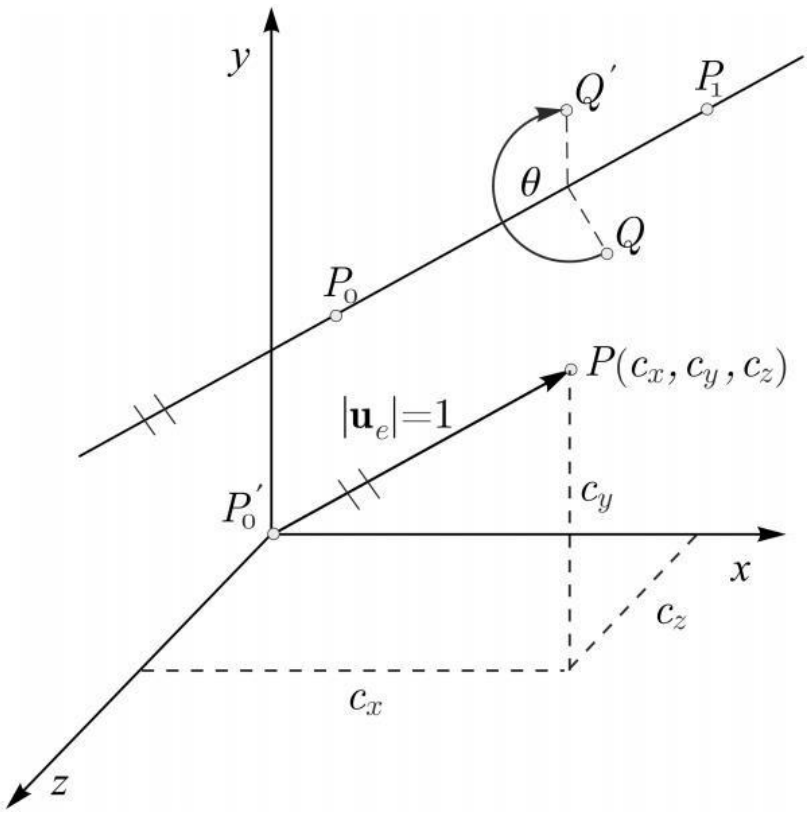


Procedure

- Translate (x_0, y_0, z_0) so that the point is at origin
- Make appropriate rotations to make the line coincide with one of the axes, say z-axis
- Rotate the object about z-axis by required angle
- Apply the inverse of step 2
- Apply the inverse of step 1

Credit: http://ami.ektf.hu/uploads/papers/finalpdf/AMI_40_from175to186.pdf | http://web.iitd.ac.in/~hegde/cad/lecture/L6_3dtrans.pdf

Direction Cosine (1/2)

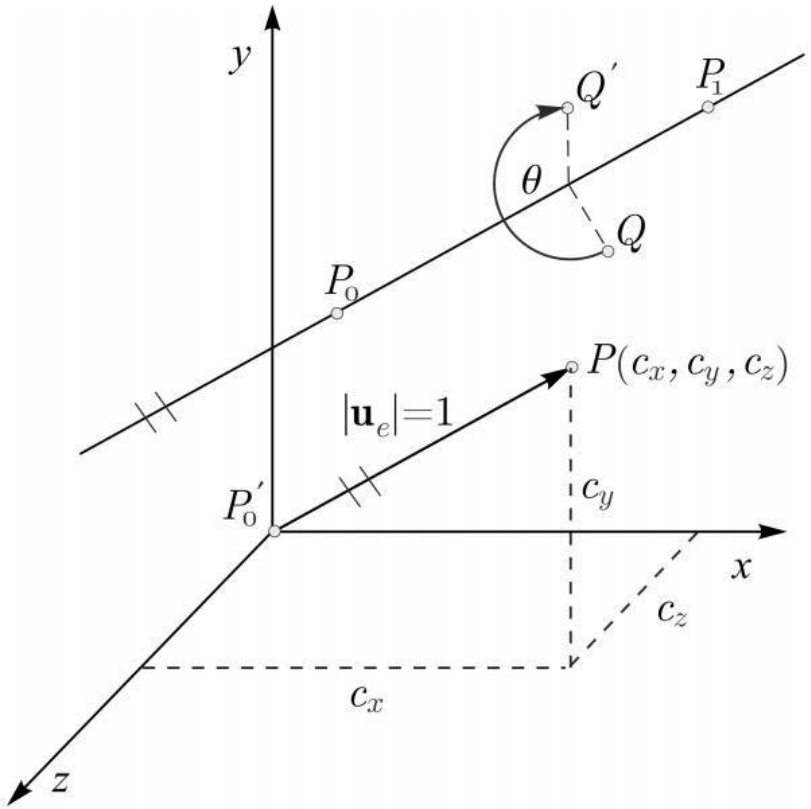


- Direction cosines of a vector are the cosines of the angles between the vector and the three coordinate axes.

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Direction Cosine (2/2)

Equivalently, they are the contributions of each component of the basis to a unit vector in that direction.



$$\mathbf{u} = P_1 - P_0$$

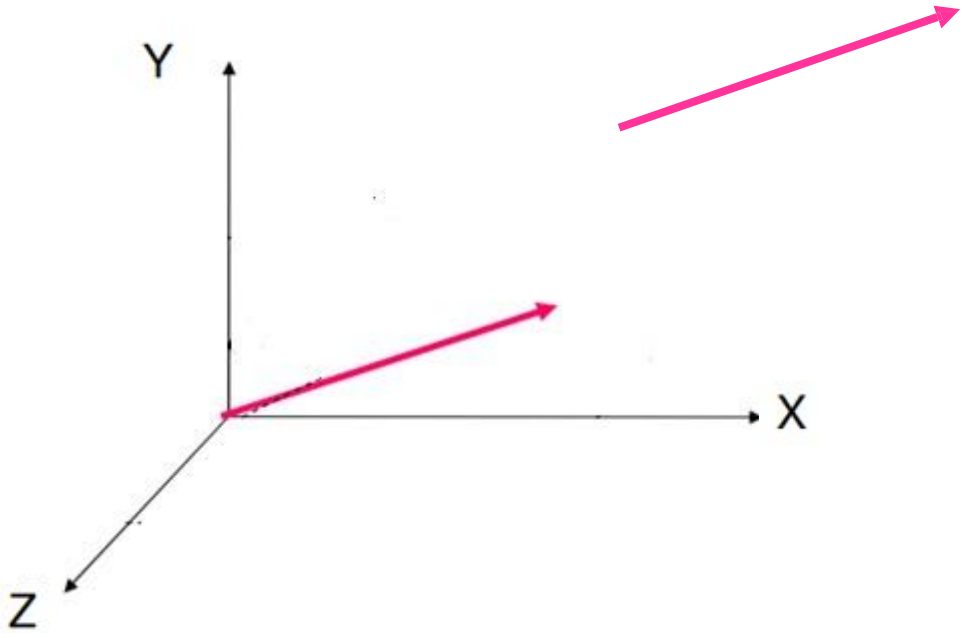
$$\mathbf{u_e} := \frac{\mathbf{u}}{|\mathbf{u}|} = (c_x, c_y, c_z)$$

$$c_x^2 + c_y^2 + c_z^2 = 1$$

$$\cos\phi_x = c_x, \quad \cos\phi_y = c_y, \quad \cos\phi_z = c_z$$

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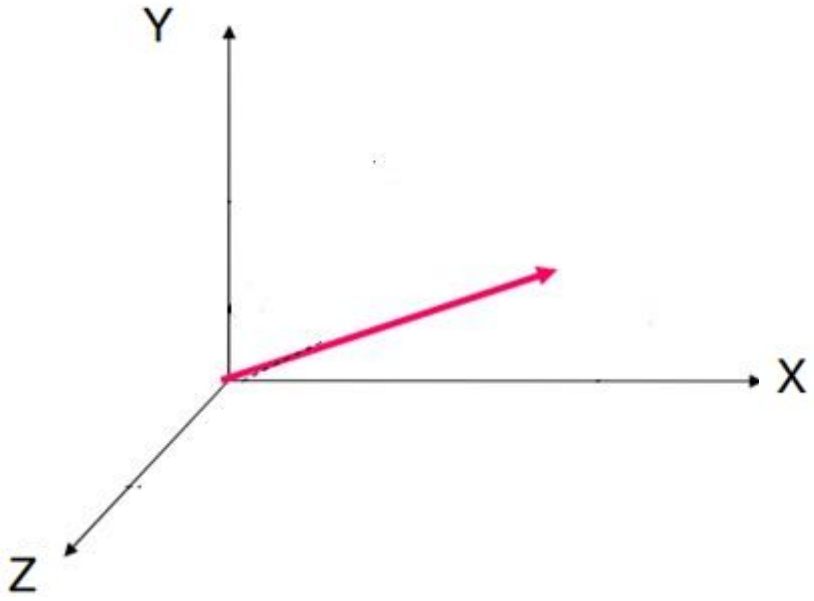
Coinciding the line with Principal axis (1/5)

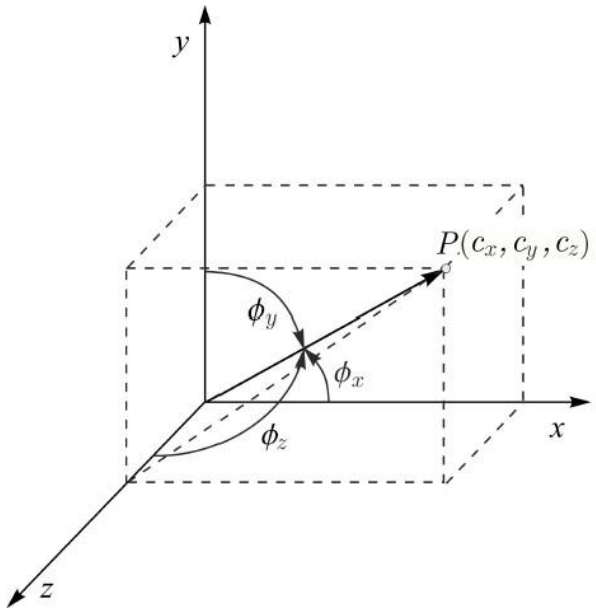


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Coinciding the line with Principal axis (2/5)

Coinciding the arbitrary axis with any axis
the rotations are needed about other two axes

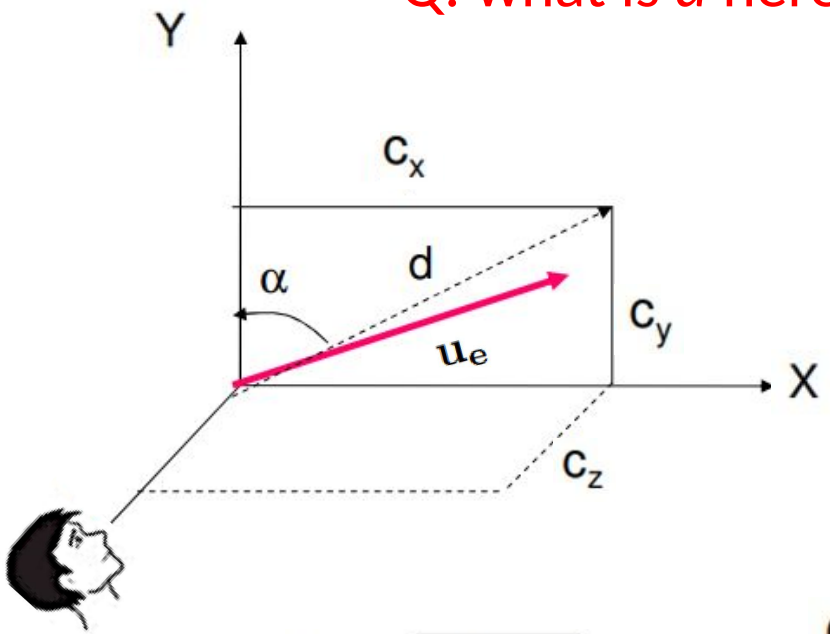


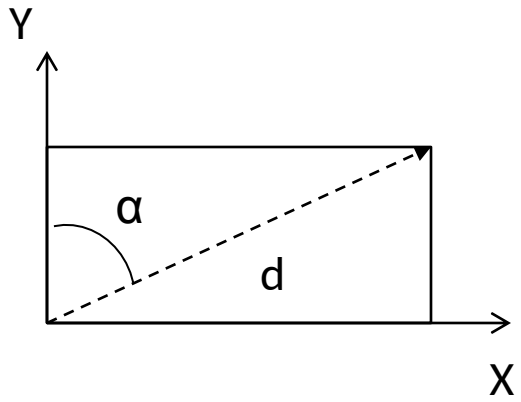


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Coinciding the line with Principal axis (3/5)

Q: what is *d* here?

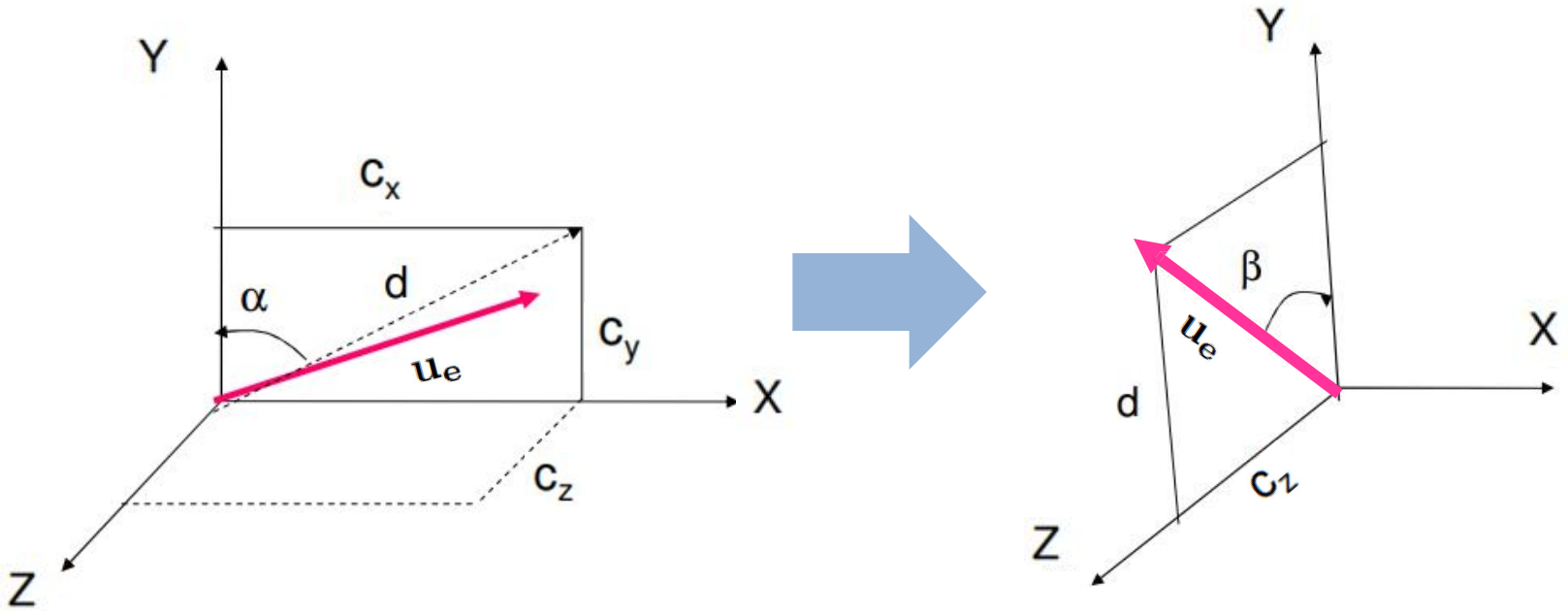




$$d = \sqrt{c_x^2 + c_y^2} \qquad \cos\alpha = \frac{c_y}{d} \qquad \sin\alpha = \frac{?}{d}$$

Source: http://web.iitd.ac.in/~hegde/cad/lecture/L6_3dtrans.pdf

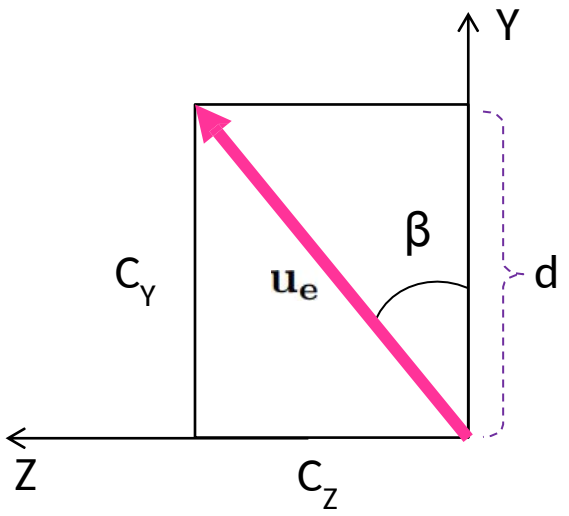
Coinciding the line with Principal axis (4/5)

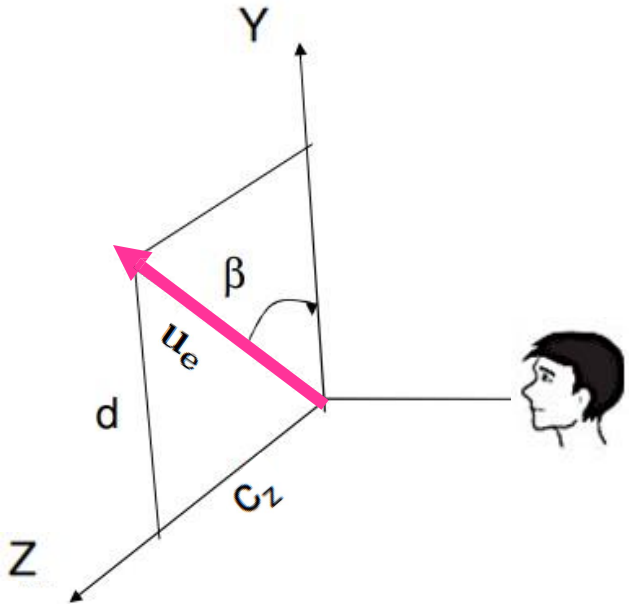


What is the rotation matrix?

Source: http://web.iitd.ac.in/~hegde/cad/lecture/L6_3dtrans.pdf

Coinciding the line with Principal axis (5/5)

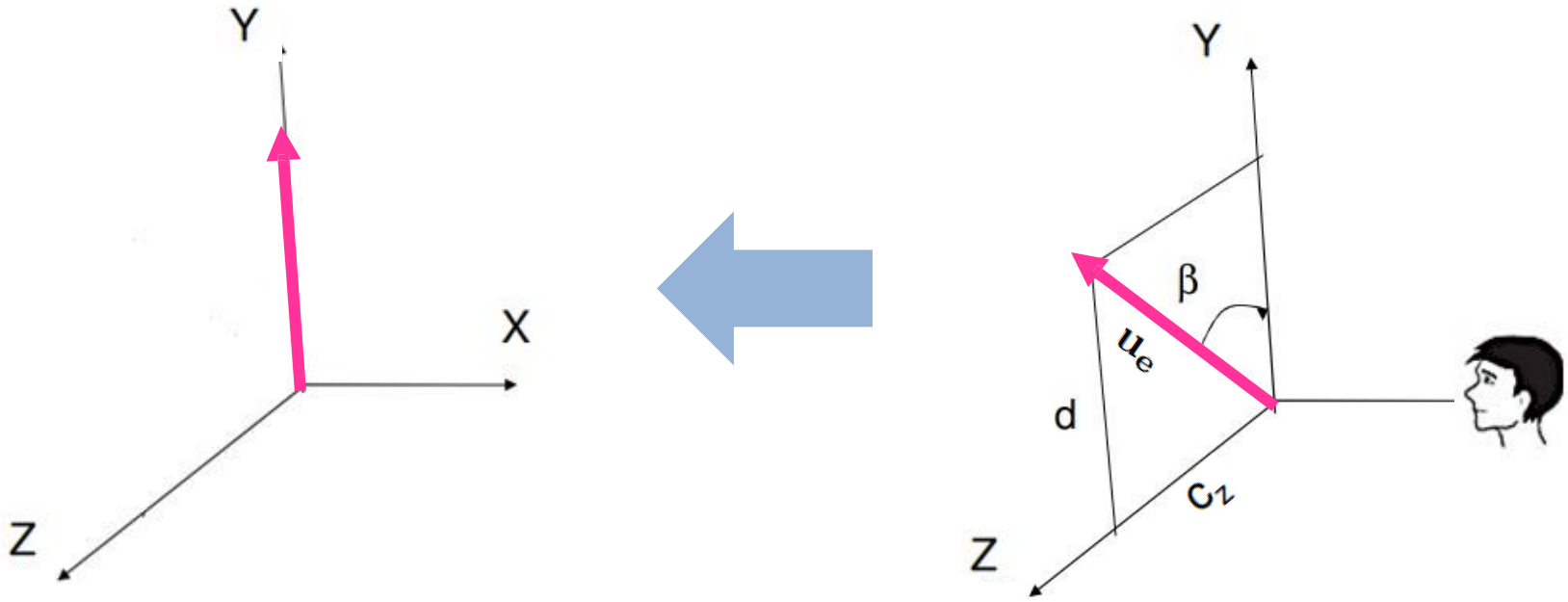




$\cos\beta = d$ $\sin\beta =$?

Source: http://web.iitd.ac.in/~hegde/cad/lecture/L6_3dtrans.pdf

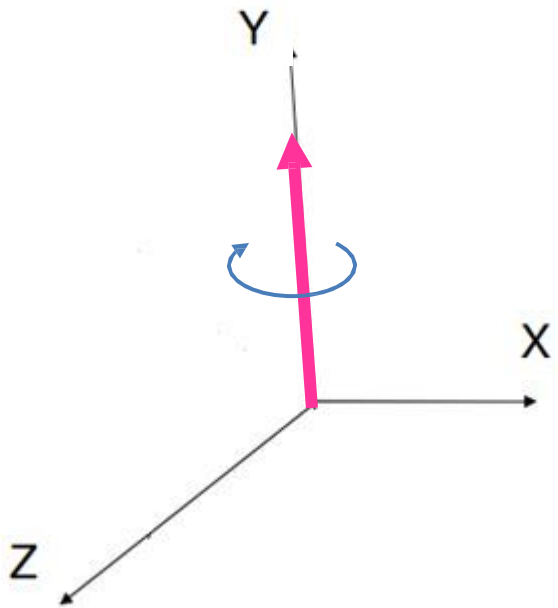
Rotating about the principal axis (1/2)



What is the rotation matrix?

Source: http://web.iitd.ac.in/~hegde/cad/lecture/L6_3dtrans.pdf

Rotating about the principal axis (2/2)



What is the rotation matrix?

Source: http://web.iitd.ac.in/~hegde/cad/lecture/L6_3dtrans.pdf

Undoing the steps (1/1)

- Q: What are the undoing steps?

Composite Transformation (1/1)

- $M = T^{-1} * R_z^{-1}(\alpha) * \ldots ?$

- $Q' = M * Q$

Practice Problem

- AB is a line and P is a point in 3D space; where the points A,B and P are (1,1,1), (3,3,4) and (2,2,4) respectively. We want to rotate P along AB by +90 degree. Determine the composite transformation matrix to do the task and calculate the rotated point P'.

References

- http://ami.ektf.hu/uploads/papers/finalpdf/AMI_40_from175to186.pdf
- http://web.iitd.ac.in/~hegde/cad/lecture/L6_3dtrans.pdf